ICLAB 2019 Spring

Paper Based Exam

Lec01(10%)

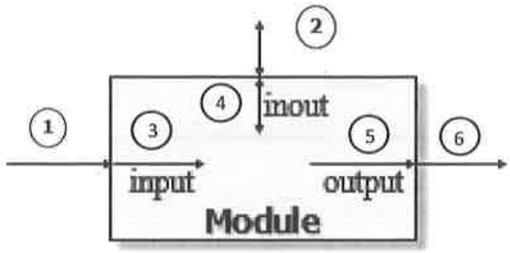
1.

a. Please clearly explain what is cell-based and full-custom design flow? (2%)

b. What is the advantage and disadvantage of them? (2%)

2.

Please write down all the possible data type below this picture. (6%)



Lec02(20%)

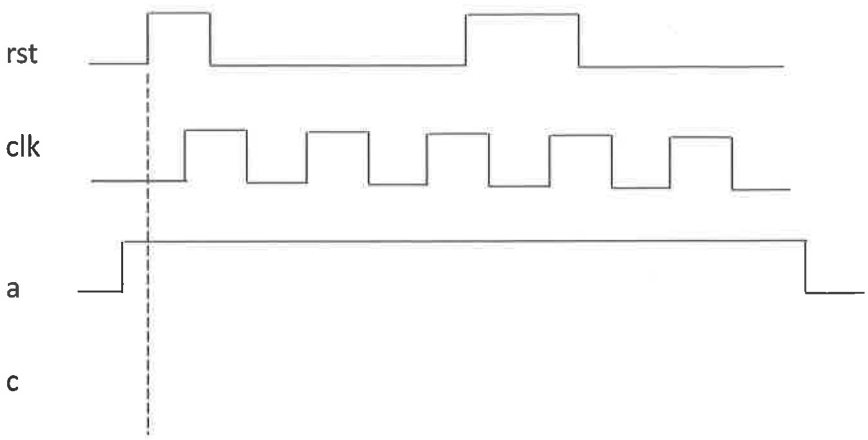
1.

Please write down the advantage and disadvantage for the synchronous and asynchronous (at least two things in each case). **(5%)**

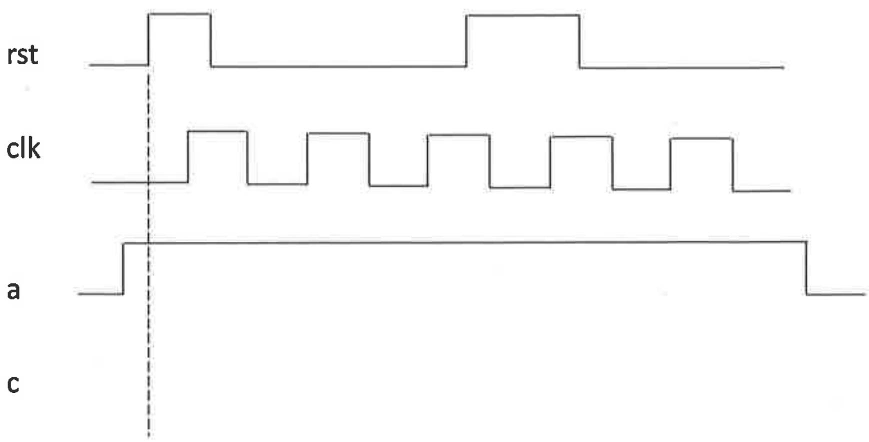
According to the provided code, finish the below waveform. (before the dotted line the c value is 2) **(2.5% + 2.5%)**

c is a five-bits signal

```
always @(posedge clk)
begin
        if(rst)   c <= 0;
        else      c <= a + 1;
end
```




```
always @(posedge clk or posedge rst)
begin
        if(rst)   c <= 0;
        else      c <= a + 1;
end
```



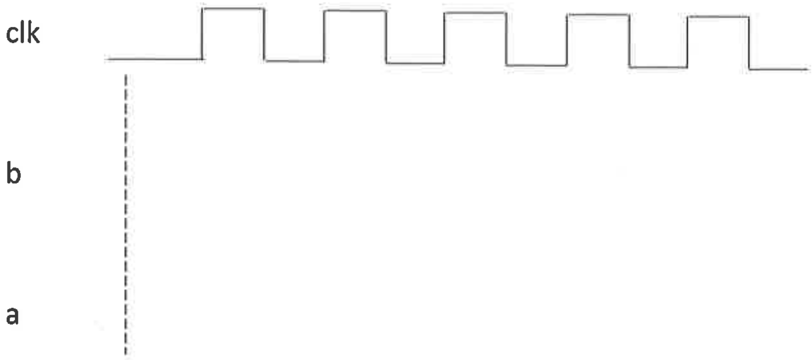
2.

According to the provided code, finish the below waveform.

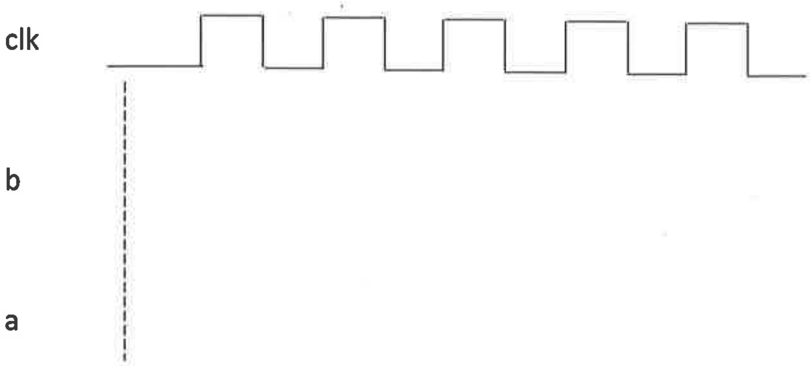
(the initial block start at the dotted line and the #1 means the very little delay)

Clock period is 10ns.**(5% + 5%)**

```
initial begin
   a=0; b=1;
end
always @(posedge clk)
begin
   b = #1 a;
   a = #1 b;
end
```



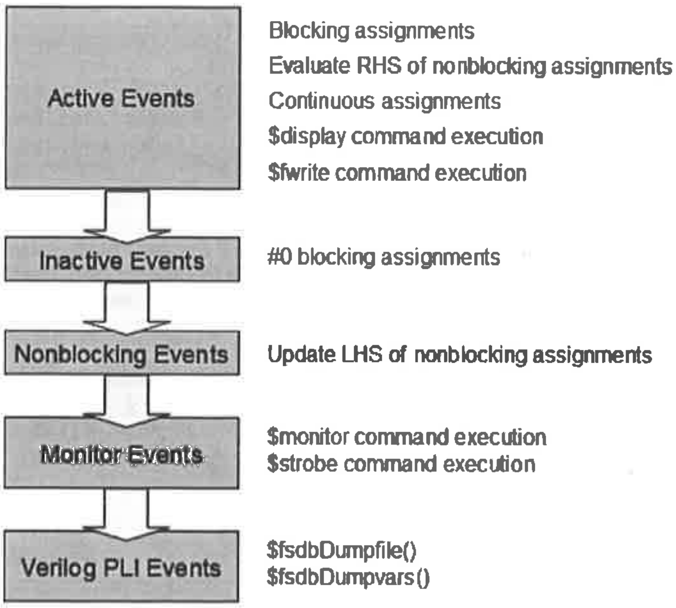

```
initial begin
      a=0; b=1;
end
always@(posedge clk)
begin
      b <= #1 a;
      a <= #1 b;
end
```



Lec03(20%)

1. According to the figure of priority below, if we wrote the program below, what will the simulation output show? Please write the result row by row separately (not only the data but also the words that show in the command.).**(10%)**

```
module priority;

reg a, b;

initial begin
   a = 0;
   b = 0;
   #1;
   a = 0;
   b = 1;
   a <= b;
   b <= a;

   $display("%0dns :\$display: a=%b b=%b"  , $stime, a, b);
   $monitor("%0dns :\$monitor: a=%b b=%b"  , $stime, a, b);
   $strobe ("%0dns :\$strobe : a=%b b=%b\n", $stime, a, b);
#0 $display("%0dns :#0       : a=%b b=%b"  , $stime, a, b);

#1 $monitor("%0dns :\$monitor: a=%b b=%b"  , $stime, a, b);
   $strobe ("%0dns :\$strobe : a=%b b=%b\n", $stime, a, b);
   $display("%0dns :\$display: a=%b b=%b"  , $stime, a, b);
   $fwrite(fp, "%0dns :\$fwrite : a=%b b=%b\n", $stime, a, b);
#0 $display("%0dns :#0       : a=%b b=%b"  , $stime, a, b);

end

endmodule
```



2. There are 6 syntax errors in the following code, please find 5 errors and correct them.**(10%)**

```
`timescale 1ns/10ps
module TESTBED;

wire in_a, in_b, in_sel;
wire rst, clk;
wire out;

initial begin
  `ifdef RTL
    $fsdbDumpfile("DESIGN.fsdb");
    $fsdbDumpvars(0,"+mda");
  `endif
  `ifdef GATE
    $fsdbDumpfile("DESIGN_SYN.fsdb");
    $fsdbDumpvars(0,"+mda");
  `endif
end

DESIGN (.clk(clk),.rst(rst),.in_a(in_a),.in_b(in_b),.in_sel(in_sel),.out(out));

PATTERN(.clk(clk),.rst(rst),.in_a(in_a),.in_b(in_b),.in_sel(in_sel),.out(out));


endmodule
```

```
module PATTERN( clk, rst, in_a, in_b, in_sel, out);

output in_a, in_b, in_sel;
output rst, clk;
input out;

reg in_a, in_b, in_sel, rst, clk;

always #(1) clk=~clk;
initial begin
    clk=0;rst=1;
    #(5)rst=0;
    in_a=0;in_b=1;in_sel=0;
    #(5)sel=1;
end

endmodule
```

```
module DESIGN( clk, rst, in_a, in_b, in_sel, out);

input in_a, in_b, in_sel;
input rst, clk;
output reg out;

reg 2_to_1;

assign out=(rst==1'b1)? 1'b0:2_to_1;
always @(posedge clk or posedge rst) begin
    mux_task;
end

task mux_task;
begin
    if(in_sel==0)    2_to_1 = in_a;
    else             2_to_1 = in_b;
end
endtask

endmodule
```

Lec04(20%)

1. Briefly describe what operator inference and instantiate IP are. What are their respective advantages and disadvantages?**(5%)**

2. (a) Briefly describe what clk-to-Q propagation delay and logic propagation delay are?**(5%)**
   (b) Give the following formulas in terms of (t_{cycle}: clock cycle、t_{setup}: setup time、t_{hold}: hold time、t_{pcq}: clk-to-Q propagation delay、t_{ccq}: clk-to-Q contamination delay、t_{pd}: logic propagation delay、t_{cd}: logic contamination delay).**(10%)**

	Setup time	Hold time
Data required time =		
Data arrival time =		
Criterion :	( ) > ( )	( ) > ( )

Lec05(15%)

1. When using memory, we use memory compiler to generate two files used in our design flow, v file and lib file. Separately explain what are this two files and which part of design flow are they used in.**(10%)**

2. Why using sequential logic to describe control signals of memory such as WEN, A, D, ..., or else will be easier to face hold time violation in gate level simulation.**(5%)**

Lec06(15%)

1. Please list three specify libraries in synthesis flow, and give a brief description of these libraries.**(7.5 %)**

2. Please explain the difference between Top-down compile and Bottom-up compile, and list their respective advantage.**(7.5%)**